J Braz. Soc. Mech. Sci. Eng. (2014) 36:737–742
DOI 10.1007/s40430-013-0124-9

TECHNICAL PAPER

A comparison of Fourier pseudospectral method and finite volume method used to solve the Burgers equation

A. A. Nascimento · F. P. Mariano · A. Silveria-Neto · E. L. M. Padilla

Received: 28 September 2013 / Accepted: 20 December 2013 / Published online: 27 February 2014
© The Brazilian Society of Mechanical Sciences and Engineering 2014

Abstract The Burger's equation serves as a useful mathematical model to be applied in fluid dynamic problems. Besides, this equation is often used to test and compare numerical techniques. In the present paper, the numerical solutions determined by both the finite volume method (FVM) and the Fourier pseudospectral method (FPM), and the results are compared with the analytical, exact solution. Furthermore, performance in obtaining the results is properly reported. The analysis includes periodic and non-periodic domains, for both methods. The non-linear term in the FVM, is discretion by using upwind and central-differencing schemes, showing a rate of convergence for the second-order accurate. Due requirement of the domains in FPM to be periodic, in the present work, the immersed boundary methodology is used, to solve the equation at non-periodic domains. It will also be shown that this approach damages the accuracy and the rate convergence of the FPM.

Technical Editor: Francisco Ricardo Cunha.

A. A. Nascimento (✉) · F. P. Mariano
Escola de Engenharia Elétrica, Mecânica e Computação, Universidade Federal de Goiás, Av. Universitária, 1488, Bloco: A, Piso: 3, Setor Leste Universitário, Goiânia, GO 74605-010, Brazil
e-mail: aanascimento@ufg.br

F. P. Mariano
e-mail: fpmariano@ufg.br

A. Silveria-Neto · E. L. M. Padilla
Faculdade de Engenharia Mecânica, Universidade Federal de Uberlândia, Av. João Naves de Ávilla, 2121, Bloco: 5P, Santa Mônica, Uberlândia, MG 38400-902, Brazil
e-mail: aristeus@mecanica.ufu.br

E. L. M. Padilla
e-mail: epadilla@mecanica.ufu.br

Keywords Burgers equation · Finite volume method · Fourier pseudospectral method · Immersed boundary method

List of symbols

U	Velocity at axis x (m/s)
t	Time (s)
x	Axis coordinate x (m)
u_b	Velocity of diffusion (m/s)
NLT	Non-linear term
Δx	Dimension of volume cell at direction x
W	West cell
P	Principal cell
E	East cell
u_P	Velocity at the main cell
u_e, u_w	Velocity at that volume face

Greek symbols

δ Distance between the central of cell

1 Introduction

Burgers in a remarkable series of papers from 1939 to 1965 [1, 11] developed a mathematical model after investigated several aspects of turbulence [2]. Its mathematical model was named Burger's equation. Effectively this equation solves problems of a fluid flow nature in which shocks and viscous dissipation is a significant factor [16].

The Burger's equation consists of partial differential equation (PDE). It is similar to one-dimensional Navier–Stokes equation without the pressure gradient term.

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2} \qquad \text{in} \quad \Omega \qquad \forall t > 0. \tag{1}$$

About the solution of the Burger's equation, we mention some works: Caldwell et al. [2] used the finite element method to allow the solution with Robin's boundary conditions. Romão et al. [15] used Eq. (1) to compare the Galerkin finite element method with the finite difference method. In this work, the Galerkin method had high accuracy solution.

Although Canuto et al. [3] affirm that the Fourier spectral method is used in problems with periodic boundary conditions, Roache [14] developed a technique to solve the problems with non-periodic boundary conditions, named pseudospectral fast Fourier transform methods. This technique is the so-called reduction to periodicity.

In the present work, the Burger's equation is solved using two methodologies, the finite volume method and the Fourier Pseudospectral method. The study comprises the comparison between the methodologies considering accuracy, convergence rate and computational cost.

2 Analytical solution

Equations (2), (3) and (4) are the analytical solution of Burgers equation used in this paper, as presented by Canuto et al. [3],

$$u(x,0) = c + u_b(x,0), \tag{2}$$

$$u_b(x,t) = -2\nu \frac{\frac{\partial \phi}{\partial x}(x-ct,t+1)}{\phi(x-ct,t+1)}, \tag{3}$$

$$\phi(x,t) = \sum_{n=-\infty}^{\infty} e^{\frac{-[x-(2n+1)\pi]^2}{4\nu t}}, \tag{4}$$

where c is advection velocity and ν is the kinematic viscosity coefficient (m^2/s).

This analytical solution is used to assess the numerical code developed, using periodic and non-periodic boundary conditions.

3 Finite volume method: FVM

The finite volume method uses the integral form of the conservation equation. The solution domain is subdivided into a finite number of small control volume control volumes which corresponds to the mesh cells as illustrated in Fig. 1.

FVM with second order of the spatial discretization, [6,13], uses the integration around the main volume. Then integrating the Eq. (1) at the time and space, it is Eq. (5):

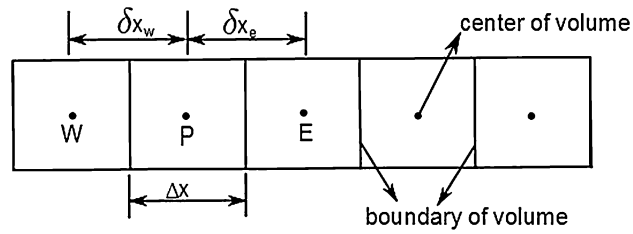

Fig. 1 Scheme of the domain divisions performed with the finite volume method

$$\frac{u^{t+\Delta t} - u^t}{\Delta t} = \int_w^e \frac{\mathrm{NLT}}{\Delta x} + \nu\left(\frac{u_E - u_P}{\delta X_e} - \frac{u_P - u_W}{\delta X_w}\right)\frac{1}{\Delta x} \tag{5}$$

where NLT represent the non-linear term.

There are several ways to solve the NLT [3]. We chose the skew-symmetric form, given by,

$$\frac{1}{2}\left[u\nabla u + \frac{1}{2}\nabla(uu)\right] \tag{6}$$

The non-linear term, integrated by the skew-symmetric form is expressed as,

$$\frac{u_P(u_e - u_w)}{2\Delta x} + \frac{\phi_e u_e - \phi_w u_w}{4\Delta x}, \tag{7}$$

where the mass flow through the faces of the control volume multiplies the velocities in these faces, ϕ_e and ϕ_w. These velocities are obtained by upwind scheme, Eq. (8), or central-difference scheme, Eq. (9) [5].

The upwind scheme approximates the velocity values at interface with the upstream value of the neighbor volume, depending of the flow direction. Thus,

$$(u_P > 0) \quad \begin{matrix}\phi_w = \phi_W \\ \phi_e = \phi_P\end{matrix} \; ; \quad (u_P < 0) \quad \begin{matrix}\phi_w = \phi_P \\ \phi_e = \phi_E\end{matrix}, \tag{8}$$

Another straightforward approximation for the value at interface is a linear interpolation between the two nearest volumes, Eq. (9). This approximation is named central-difference scheme.

$$\phi_e = \frac{\phi_E - \phi_P}{2}; \quad \phi_w = \frac{\phi_P - \phi_W}{2}. \tag{9}$$

4 Fourier pseudospectral method: FPM

The solution of the Burger's equation with the FPM uses the discrete Fourier transform. The FPM is applied at Eq. (1) and can be rewritten as:

$$\frac{d\hat{u}}{dt} = -ik\hat{u} \times \hat{u} + \nu(ik)^2\hat{u}, \tag{10}$$

where '^' is the variable transformed to Fourier space, k is the wave number and $i = \sqrt{-1}$.

It can be noted the convolution product at NLT that requires high computation cost to solve. Then, to avoid it, Canuto et al. [3] propose the Fourier Pseudospectral method, that consists in first calculate the divergent form, following the steps:

1. Calculate the multiplicative of the velocity at physical space, as uu;
2. Transform the product to spectral space, as uu;
3. Calculate the derivative and multiply by 1/2 after calculate the advective form,
4. Transform the field u to spectral space;
5. Calculate the derivatives, $ik\hat{u}$;
6. Calculate the inverse transformed of the velocity derivatives multiply by the velocity in the physical space.
7. Transform the derivative from the physical space to spectral space,

Equation (11) shows discretized form using Euler first order in time. However, the results were obtained using the classical fourth-order Runge–Kutta method, as proposed by Canuto et al. [4].

5 Immersed boundary method: IBM

The immersed boundary method was developed by Peskin [12] with the aim of simulated the cardiac mechanics valves and associated blood flow. This promisor method is used in the present paper to enable solution of the non-periodic problem using the FPM. First, it is defined a domain, with nodes uniformly distributed named Eulerian domain (Ω). Next, it is defined as the Lagrangian domain (Γ), where the force acts [9]. We consider that the Lagrangian nodes are coincident with the Eulerian nodes, as showed in Fig. 2. In this figure, the full line represents the Eulerian domain and the symbols are the nodes of Lagrangian domain.

The spectral method works only periodic boundary conditions. For the solution at non-periodic domain, the Eq. (1) was modified by Eq. (11).

$$\frac{u^{t+\Delta t} - u^t}{\Delta t} = -u\frac{\mathrm{d}u}{\mathrm{d}x} + \nu\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + f_x \tag{11}$$

where f_x is the force term, which represents the boundary condition according to the IBM. For coincident nodes,

$$\begin{aligned} f_x &= F_x(X,t) \quad \text{if} \quad \vec{x} = \vec{X} \\ f_x &= 0 \quad \text{if} \quad \vec{x} \neq \vec{X} \end{aligned}, \tag{12}$$

where $\vec{x}$ represents the position of any point at the Ω and position $\vec{X}$ any point at the Γ (see Fig. 2).

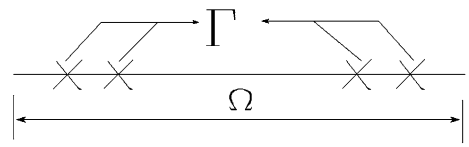

Fig. 2 Sketch of domain used by the immersed boundary method

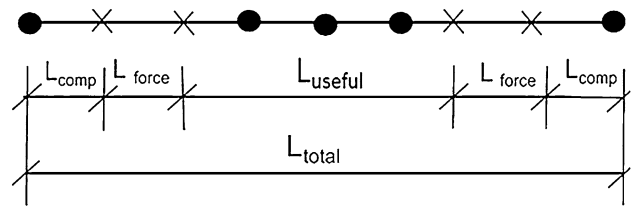

Fig. 3 Sketch of immersed boundary method, used in this work

The direct-forcing method (DFM), developed by Mohd-Yusof [10], is used to calculate the Lagrangian fields. This method obtains the force directly of the numerical solution that is determined by difference between values of the solved and physical velocities. Equations (13) and (14) show this concept in discretized form.

$$\frac{u^{t+\Delta t} - u^* + u^* - u^t}{\Delta t} - \text{rhs} - f_x = 0\ , \tag{13}$$

$$\frac{u^* - u^t}{\Delta t} - \text{rhs} = 0\ , \tag{14}$$

where rhs groups the convective and viscous terms at any time level between t and $t + \Delta t$. The force term is that generates the desired velocity u^*. The Lagrangian force term is given by Eq. (15) over X_l.

$$F_x = \frac{U^{t+\Delta t} - U^*}{\Delta t} \quad \forall\, X_l \tag{15}$$

where $U^{t+\Delta t}$ is the physical boundary condition and U^* is the temporal parameter. Lastly, the update of the Eulerian velocity is given by Eq. (16), where f_x is valued by Eq. (12).

$$u^{t+\Delta t} = u^* + f_x \Delta t\ . \tag{16}$$

Figure 3 represents the one-dimension domain used to study the Burgers equation for non-periodic boundary condition. The L_{total} is the complete full Eulerian domain, which is composed of L_{force}, the forcing zone, L_{useful}, the useful domain and L_{comp}, the complementary domain.

6 Results

The simulation was realized using 32, 64, 128 and 256 grids (volumes and collocation points/nodes), advection velocity c is 4.0 m/s and kinematic viscosity 0.2 m^2/s. The

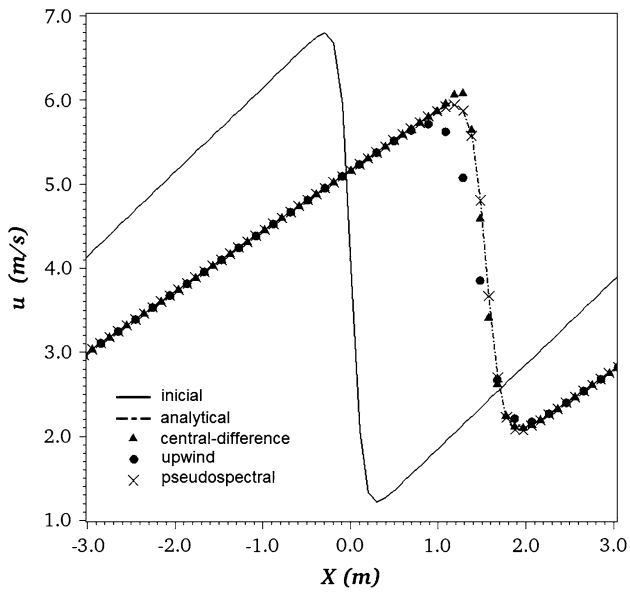

Fig. 4 Solution of Burger's equation using 64 grid points, simulations of periodic domain

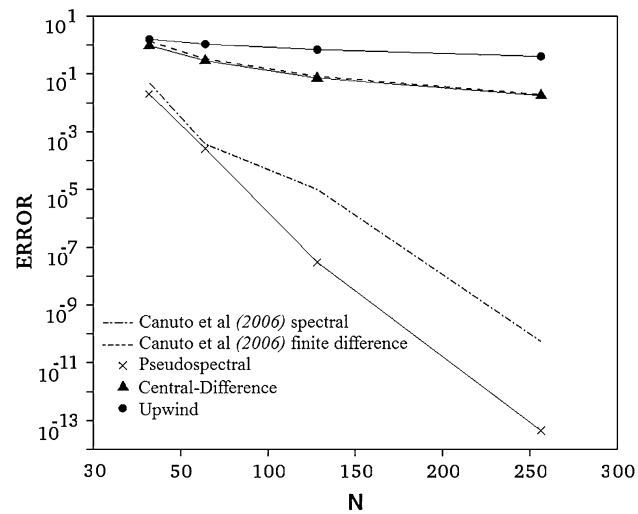

Fig. 5 Maximum error, in $t = \pi/8$ (s) to different grid points

simulations for the non-periodic cases were performed using 2, 4 and 6 points at forcing zone.

6.1 Periodic domain

Figure 4 shows the initial condition and Burgers equation solution at time $^{\pi}/_{8}$ (s), using grid with 64 volumes/points. In this time, the numerical differences can be observed with the analytical solution.

The solution of the Burger's equation at initial condition, as well as, the solution at time $\pi/8$ (s) is shown in Fig. 4. These results were obtained with a grid with 64 volumes/nodes. For time $\pi/8$ (s), it can be observed the numerical differences with the analytical solution. As expected, the approach using FVM with upwind scheme presents results with more numerical diffusion and the results using central-difference approach that are better than previous, but presents over-prediction at the discontinuity region. The results that are similar with the analytical solution are obtained using FPM.

The high accuracy of the spectral method for the discontinuous problem and periodic boundary conditions are evidenced in Fig. 5. This figure shows that refining the grid with FPM, the maximum error tends to be 10^{-15}, approaching the results obtained by Canuto et al. [4]. The solution using FVM with upwind scheme tends to be 10^{-1} and that using the central-difference scheme tends 10^{-2}. On the other hand, the errors using FVM with central-difference scheme coincide with the results obtained by Canuto et al. [4] for finite difference method.

The data in Table 1 show the computational cost, in seconds, of all simulations performed with periodic boundary conditions. These data were obtained using a processor Core Due of 2.66 Ghz. In this table, N is the grid points, FVM-UP is the finite volume method with scheme upwind and FVM-CD is the finite volume method using central-difference scheme.

Table 1 Computational cost to periodic domain

N	FVM-UP	FVM-CD	FPM
32	2.399×10^{-2}	2.500×10^{-2}	9.099×10^{-2}
64	8.899×10^{-2}	8.899×10^{-2}	0.3290
128	0.7040	0.7180	2.581
256	5.530	5.7470	20.73

The computational cost to FVM has little difference between them, but the FPM is three times more than FVM. The high computational cost at FPM can be due the number of times that fast Fourier transform (FFT) is requested. However, the accuracy of FPM is approximately seven times more than FVM.

6.2 Non-periodic domain

The several solutions for the Burgers equation in a non-periodic domain are presented in Fig. 6, with a grid 64 volumes/points. As mentioned before, to solve this problem with the FPM was used the immersed boundary method with DFM, as suggested by Mariano et al. [8] and Mariano [7]. This figure shows the non-periodic solutions for both methodologies considered. The result for FPM shows only the useful domain.

The spatial convergence rates of the methodologies are depicted in Fig. 7. In Fig. 7a, the convergence rates using the pseudospectral method were obtained with application of the IBM using 1, 2, and 3 points on each side of the

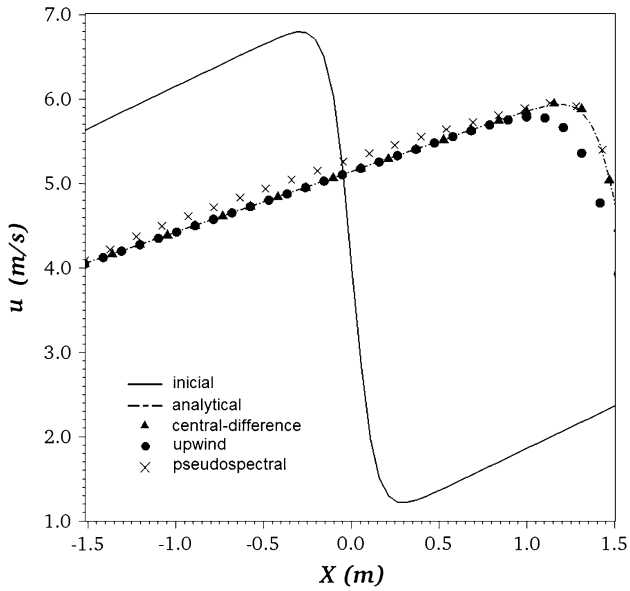

Fig. 6 Solution of Burger's equation using 64 grid points, simulations of non-periodic domain

forcing zone L_force. We see that the increase in the points in the forcing zone changes the order of convergence that tends to be second order, where the accuracy is better. It is possible because increasing the number of points ensures the improvement of non-periodic boundary condition. Results of Fig. 7b only confirm that upwind scheme is first convergence order and central-difference is second convergence order, even for non-periodic domain.

Observing Fig. 7, we can see that the convergence and accuracy of the FVM using central-difference scheme have better result than for FPM using six points in the forcing zone. So, the use of the immersed boundary method decreases the convergence rate and accuracy of the FPM.

Computational cost of simulations for the problem with non-periodic domain, with all meshes and 1, 2 and 3 points in each side of useful domain for FPM, are shown in Table 2. These values were obtained using the same processor. The coupling of the FPM with the IBM, to solve the non-periodic domain, increases the computational cost in approximately five times when compared with the computational cost for FPM with periodic domain. Using the three points in the forcing zone, the computational cost increases approximately 1.0 % when compared with one point.

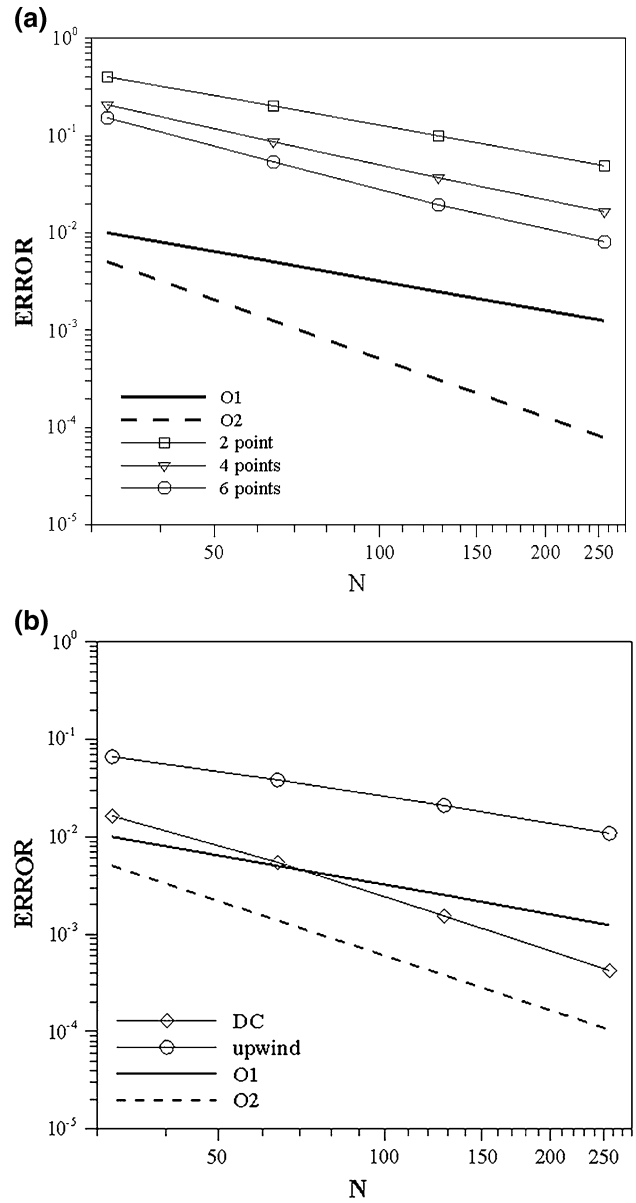

Fig. 7 Convergence rate, at the time $t = \pi/8$ (s). **a** Pseudospectral method using IBM, **b** finite volume method

Table 2 Computational cost to non-periodic domain

N	1 points	2 points	3 points
32	0.200	0.207	0.204
64	1.379	1.382	1.392
128	11.89	12.09	12.15
256	100.64	101.26	101.6

7 Conclusions

The central-difference interpolation scheme used at the finite volume method showed better than upwind results, which present high numerical dissipation. Therefore, the central-difference interpolation scheme improves the order of finite volume method.

As reported in the present paper, it was possible to conclude that Fourier spectral method is well suited to the

calculation of problems with periodic boundary conditions. Although, the FPM has higher computational cost when compared with the FVM. On the other hand, the coupling of the FPM with the IBM, to solve the non-periodic domain, causes the decrease of accuracy, as well as, causes decrease of convergence rate to second order. Nevertheless, FPM coupled to IBM has high accuracy yet.

Is importantly that FPM coupled to IBM enables to solve non-periodic problems, which justifies your use.

Acknowledgments The authors would like to thank the CAPES, CNPq, FAPEMIG, FAPEG and PETROBRAS for the financial support. To the School of Mechanical Engineering of the Federal University of Uberlândia and to the School of Electrical, Mechanical and Computation Engineering of the Federal University of Goiás.

References

1. Burger JM (1974) The nonlinear diffusion equation. D. Reidel Publishing Company, Dordrecht-Boston
2. Caldwell J, Wanless P, Cook AE (1981) A finite element approach to Burger's equation. Appl Math Model 5:189–193
3. Canuto C, Quarteroni A, Hussaini MY, Zang T (1988) Spectral methods in fluid dynamics, 2nd edn. Springer, New York
4. Canuto C, Quarteroni A, Hussaini M, Zang TA (2006) Spectral methods fundamentals in single domains. Spring, New York
5. Ferziger JH, Peric M (2002) Computational methods for fluid dynamics. Springer, New York
6. Maliska CR (1995) Transferência de calor e Mecânica dos Fluidos computacional. Livros Técnicos Científicos Editora S. A. LTC, São Paulo
7. Mariano FP (2011) Soluções Numéricas de Navier Stokes Utilizando uma hibridação das Metodologias Fronteira Imersa e Pseudospectral de Fourier. Universidade Federal de Uberlândia—Faculdade de Engenharia Mecânica, Uberlândia
8. Mariano FP et al (2010) A new incompressible Navier–Stokes solver combining Fourier pseudo-spectral and immersed boundary methods. Computer Model Eng Sci 59:181–216
9. Mittal R, Iaccarino G (2005) Immersed boundary method. Annu Rev Fluids Mech 37: 239–261
10. Mohd-Yusof J (1997) Combined immersed boundary/b-spline methods for simulation of flow in complex geometries. CTR Annu Res Breifs 1:317–327
11. Nieuwstadt FTM, Steketee JA (1995) Selected papers of J.M. Burgers. Springer, Delf
12. Patankar SV (1980) Numerical heat transfer and fluid flow. Taylor e FranciBook LTD, USA
13. Peskin C (1970) Flow patterns around heart valves: a numerical method. J Comput Phys 10:252–271
14. Roache PJ (1978) A pseudospectral FFT technique for non-periodic problems. J Comput 27:204–220
15. Romão EC, Campos MD, Moura LFM (2010) Galerkin finite element method and finite difference method for solving convective non-linear equation. Therm Eng XIX(1–2):69–73
16. Smith LC (1997) Finite element approximations of Burgers' equation with Robin's boundary conditions. Virginia Polytechnic Institute and State University, Blacksburg